FYS4411/9411 March 6

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Newton-Raphson's method

$$F(\hat{\theta}) = F(\theta^{(n)}) +$$

$$\left\{ \Delta\theta = \hat{\theta} - \theta^{(n)} \quad \left[\hat{\theta} = \theta^{(n+1)} \right] \right\}$$

$$+ \left(\vec{\nabla}_\theta F(\theta^{(n)}) \right)^T \left(\hat{\theta} - \theta^{(n)} \right)$$

$$+ \frac{1}{2} \left(\hat{\theta} - \theta^{(n)} \right)^T H(\theta^{(n)}) \left(\hat{\theta} - \theta^{(n)} \right)$$

$$+ O\left(\left(\hat{\theta} - \theta^{(n)} \right)^3 \right)$$

Truncating at second derivative

$$\hat{\theta} = \theta^{(m+1)} = \theta^{(m)} - \eta^{(m)} \nabla_\theta F(\theta^{(m)})$$

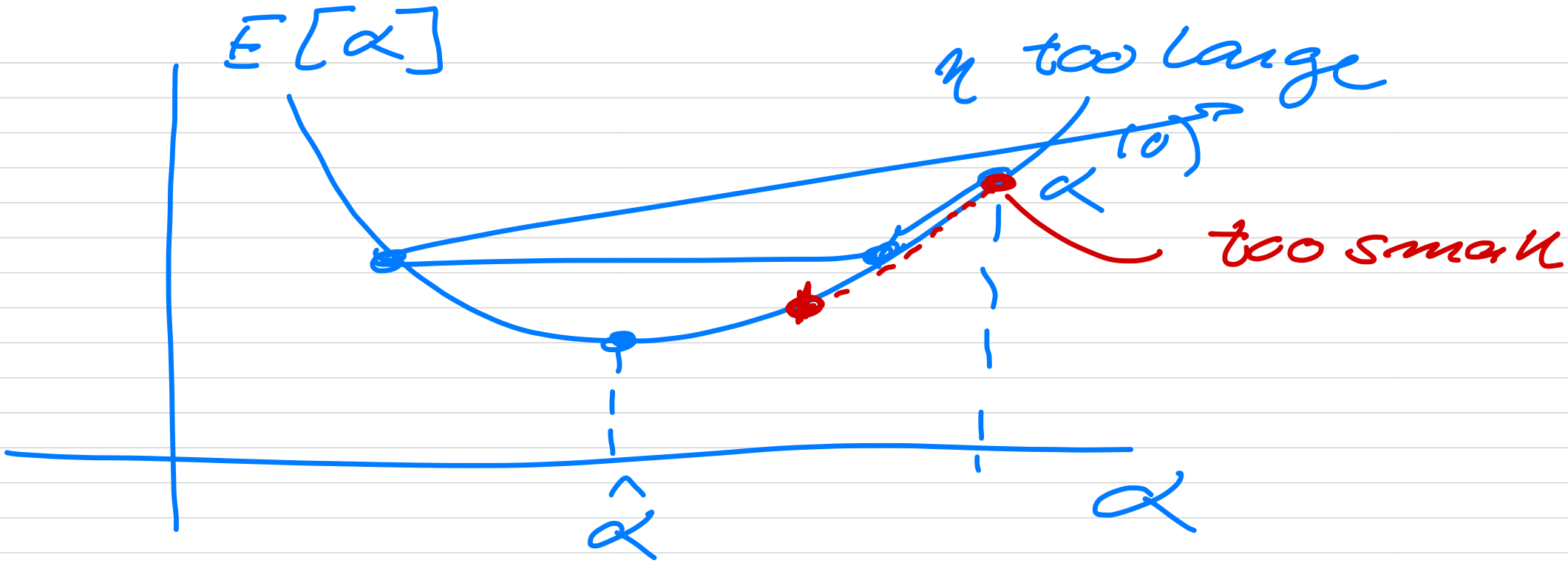
$$\eta^{(m)} = H^{-1}(\theta^{(m)})$$

* Plain gradient descent

η = constant $[10^{-4}, 10^{-3}, \ldots 10^{-1}]$

$\eta^{(m)} = \eta$

updates $\left\{ \begin{array}{l} \text{momentum} \\ \text{RMSprop, Adagrad} \\ \text{ADAMS} \ldots \end{array} \right.$



$$\|\theta^{(n+1)} - \theta^{(n)}\|_2 \le \varepsilon, \text{ Stop}$$

10^{-4}

$E[\alpha]$ and $\nabla_\alpha E[\alpha]$ $\quad 10^{-5}$ ..

with ϵ being small

($\epsilon \in \mathbb{R}^p$ p is small)

* quasi-Newton methods

BFGS (Broyden Powell ...)

use secant rule to approximate derivatives

$$f(x) = 0 \; : \; x^{(n+1)} = x^{(n)} - \frac{f(x^{(n)})}{f'(x^{(n)})}$$

Derivative

$$f'(x^{(n+1)}) \simeq \frac{f(x^{(n+1)}) - f(x^{(n)})}{x^{(n+1)} - x^{(n)}}$$

* Stochastic reconfig

recommended for PZ and VMC with NN SGD with ADAMS or similar.

Broyden's method

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$s_k = x_{k+1} - x_k$$

$$y_k = f(x_{k+1}) - f(x_k)$$

$$B_{k+1} = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k} = \frac{y_k}{s_k}$$

$$\Delta B = B_{k+1} - B_k \quad \Big| \quad B_{k+1} s_k = y_k$$

we want

$$B_{k+1} s_k = y_k \quad \text{and}$$

$$\Delta B = B_{k+1} - B_k \quad \text{is small}$$

$$B_{k+1} s_k = (B_k + \Delta B) s_k = y_k$$

$$\text{s.t.} \quad \min_{\Delta B} \|\Delta B\|_F^2$$

$$\Delta B s_k = (B_{k+1} - B_k) s_k$$

$$= y_k - B_k s_k = r_k$$

Minimize

$$\|\Delta B\|_F^2 = \mathrm{Tr}(\Delta B^T \Delta B)$$

s.t. $\Delta B s_k = r_k$

Digression: Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2}$$

$A \in \mathbb{R}^{m \times n}$

$$\|A\|_F^2 = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2$$

$$(A^T A)_{jk} = \sum_{i=1}^{m} a_{ij} a_{ik}$$

$$\mathrm{Tr}(A^T A) = \sum_{j=1}^{n} (A^T A)_{jj}$$

$$(A^T A)_{jj} = \sum_{i=1}^{m} a_{ij}^2$$

$$\mathrm{Tr}(A^T A) = \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij}^2$$

$$= \|A\|_F^2$$

Add Lagrangian multiplier

$$\mathcal{L}(\Delta B, \lambda) = \mathrm{Tr}(\Delta B^T \Delta B)$$

$$+ 2\lambda^T (\Delta B s_k - r_k)$$

$$\frac{\partial \mathcal{L}}{\partial \Delta B} = 2\Delta B + 2\lambda s_k^T \Rightarrow$$

$$\Delta B = -\lambda s_k^T$$

impose constraint

$$\Delta B s_k = -\lambda s_k^T s_k = r_k$$

$$\lambda = -\frac{r_k}{s_k^T s_k}$$

$$\Delta B = \frac{r_k s_k^T}{s_k^T s_k}$$

$$B_{k+1} = B_k + \Delta B$$

$$= B_k + \frac{(y_k - B_k s_k) s_k^T}{s_k^T s_k}$$

BFGS

$$\vec{\nabla} f(\vec{x}) = 0$$

$$x_{k+1} = x_k - H_k^{-1} \nabla f(x_k)$$

$$\left(\alpha \left(\nabla^2 f(x_k)\right)\right)^{-1}$$

$$B_{k+1} = \frac{\nabla f(x_{k+1}) - \nabla f(x_k)}{x_{k+1} - x_k}$$

$$y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$$

$$s_k = x_{k+1} - x_k$$

we want the inverse

$$x_{k+1} = x_k - H_k^{-1} \nabla f(x_k)$$

$$B_{k+1} s_k = y_k \Rightarrow$$

$$H_{k+1} y_k = s_k$$

BFGS

inverting the equations

$$\boxed{B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k}}$$